



Installation instructions

1. For Windows

- (a) Install Git: Go [here](#), download and follow the installation instructions.
- (b) Install Paraview: Go [here](#), download and follow the installation instructions.
- (c) Install Docker Desktop
 - i. Go [here](#), download and follow the installation instructions
 - ii. Start Docker desktop.
- (d) Download 
 - i. Open Terminal
 - ii. Type (on a network with no firewalls/restrictions)

```
mkdir C:\path\where\to\install
cd C:\path\where\to\install
git clone https://github.com/Michele-s-team/irene.git
```

- (e) Create the virtual environment

```
cd C:\path\where\to\install
docker pull michelecastellana/irene
```

- (f) Start the virtual environment and enter it:

```
docker run -ti --name irene_1 `
-v "C:/path/where/to/install/irene:/home/fenics/shared" `
michelecastellana/irene:latest
```

- (g) Make a test run:
 - i. Generate the mesh:

```
cd /home/fenics/shared/generate_mesh/2d/ring/
PARAMETERS_PATH="/home/fenics/shared/generate_mesh/2d/ring"
SOLUTION_PATH="$PARAMETERS_PATH/solution"
rm -rf "$SOLUTION_PATH"; mkdir "$SOLUTION_PATH"
python3 generate_mesh.py "$PARAMETERS_PATH" "$SOLUTION_PATH"
```

ii. Solve the variational problem:

```
cd /home/fenics/shared/steady_state/no_flow
MESH_PATH="/home/fenics/shared/generate_mesh/2d/ring/solution"
SOLUTION_PATH="/home/fenics/shared/steady_state/no_flow/solution"
rm -rf "$SOLUTION_PATH"
python3 solve.py ring "$MESH_PATH" "$SOLUTION_PATH"
```

The expected output is:

```
Reading parameters from /home/fenics/shared/generate_mesh/2d/ring/solution/mesh_metadata.csv...
... close.
[...]
Solving nonlinear variational problem.
  Newton iteration 0: r (abs) = 2.156e+02 (tol = 1.000e-06) r (rel) = 1.000e+00 (tol = 1.000e-06)
  Newton iteration 1: r (abs) = 1.705e+01 (tol = 1.000e-06) r (rel) = 7.908e-02 (tol = 1.000e-06)
  Newton iteration 2: r (abs) = 9.820e-01 (tol = 1.000e-06) r (rel) = 4.555e-03 (tol = 1.000e-06)
  Newton iteration 3: r (abs) = 4.952e-02 (tol = 1.000e-06) r (rel) = 2.297e-04 (tol = 1.000e-06)
  Newton iteration 4: r (abs) = 2.477e-03 (tol = 1.000e-06) r (rel) = 1.149e-05 (tol = 1.000e-06)
  Newton iteration 5: r (abs) = 1.239e-04 (tol = 1.000e-06) r (rel) = 5.744e-07 (tol = 1.000e-06)
  Newton solver finished in 5 iterations and 5 linear solver iterations.
Solving nonlinear variational problem.
  Newton iteration 0: r (abs) = 4.063e-02 (tol = 1.000e-10) r (rel) = 1.000e+00 (tol = 1.000e-09)
  Newton iteration 1: r (abs) = 8.400e-18 (tol = 1.000e-10) r (rel) = 2.067e-16 (tol = 1.000e-09)
  Newton solver finished in 1 iterations and 1 linear solver iterations.
Writing parameters to /home/fenics/shared/steady_state/no_flow/solution/metadata.csv...
... done.
[...]
F_kappa_{ds_r} = [0.00042879990256965497, 0.0014179372428020193, -5.210240236637924]
```

2. For Linux

(a) Install Git:

- i. Open Terminal
- ii. Type

```
sudo apt-get install git
```

(b) Install Paraview:

```
sudo apt-get update
sudo apt-get install paraview
```

(c) Install Docker Desktop: go [here](#), download Docker Desktop and follow the installation instructions.

(d) Download [Irene](#):

i. Open Terminal

ii.

```
mkdir /path/where/to/install/
cd /path/where/to/install/
```

iii. Run (on a network with no firewalls/restrictions):

```
sudo git clone https://github.com/Michele-s-team/irene.git
sudo chmod -R 777 irene
```

(e) Create the virtual environment:

i. Open Terminal

ii.

```
cd /path/where/to/install/
sudo docker pull michelecastellana/irene
sudo apt install -y qemu-user-static
sudo docker run --privileged --rm tonistiigi/binfmt --install all
```

(f) Start the virtual environment and enter it

```
sudo docker run -ti -v $(pwd)/irene:/home/fenics/shared \
--name irene_1 michelecastellana/irene:latest
```

(g) Make a test run with Case [1g](#).

3. For MacOS

(a) Install Git:

i. Open Terminal

ii. Type

```
/bin/bash -c \
"$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
/opt/homebrew/bin/brew install git
```

(b) Install Paraview: go [here](#), download and follow the installation instructions

(c) Install Docker Desktop

- i. Go [here](#), download and follow the installation instructions
- ii. Start Docker desktop.

(d) Download :

i. Open Terminal

- ii.

```
mkdir /path/where/to/install/  
cd /path/where/to/install/
```

iii. Run (on a network with no firewalls/restrictions)

```
git clone https://github.com/Michele-s-team/irene.git  
sudo chmod -R 777 irene
```

(e) Create the virtual environment:

i. Open Terminal

- ii.

```
cd /path/where/to/install/  
sudo docker pull michelecastellana/irene
```

(f) Start the virtual environment and enter it

```
docker run -ti \  
-v $(pwd)/irene:/home/fenics/shared \  
--name irene_1 michelecastellana/irene
```

(g) Make a test run with Case [1g](#).